Horizon — runtime pipeline (one request)
indirect injection = Stage 1 + Stage 2 (Horizon math) · direct injection / jailbreak = rose layer

Stage 1 — keyed spotlighting (offline, Slides 11–27)

Data
BIPIA clean docs · LLMail-Inject + BIPIA canary attacks

8 defenses τ, fresh key κ
none · delimit · ^ · dyn. marker N^K · ROT13
shift 25 · scramble |Σ|! · Base64

Forging
$q / |K_\tau|$ (log-space)

ASR bound
Clopper–Pearson $\text{Beta}^{-1}(1-\alpha; s+1, n-s)$

χ²-DRO
$\bar{p} + \sqrt{(2\rho\ \bar{p}(1-\bar{p}))}$

Adaptive attacker
$\text{argmax}_{\delta \in \Delta_B} E_K[A] \to \bar{p}^{adp}$

Combined risk
$\bar{R}_\tau(\rho) = \min\{1, q/|K| + \max(\text{DRO}, \text{adaptive})\}$

Choice rule
$\tau^* = \text{argmin}\ \bar{R}$ s.t. $U \geq U_{none} - \eta$

Sion game, level 1
V_1 SOCP on paired $A[x, \tau] \to \pi^*$

Stage 1 outputs
$\pi^* \cdot |K| = \min |K_\tau| \cdot \bar{R}_Q = \Sigma\ \pi^*\ \bar{R}_\tau$

User prompt u
trusted channel

Input guard
DeBERTa injection classifier (ProtectAI)

Privileged LLM
sees ONLY u → writes a program

External data
emails · web · tool outputs (untrusted)

CaMeL interpreter
tracks data + control-flow dependencies

π*, |K|

untrusted x

Quarantined LLM × m
each run: own τ ~ π*, own fresh κ
no tools · typed JSON schema

$\bar{R}Q$

Forging
$\Sigma\ \pi^* \cdot q / |K_\tau|$

QIF leakage
$\leq \Sigma \log_2|S|$ bits · cap per config

Majority vote
> m/2 agree, else refuse · score s(x) = agreement

Hoeffding
$\exp(-2m(\frac{1}{2} - \bar{R}_Q)^2)$ · vs empirical

Typed value + capability
v = (data, type, cap) · cap ∈ {SYSTEM ⊑ USER ⊑ UNTRUSTED}

Policy check
YAML rule table per tool + role rules

Noninterference
Z3 proof at startup: UNTRUSTED ⇒ ¬ALLOW

ALLOW
policy met

CONFIRM → ask user
only if s(x) ≥ λ̂, else DENY

Conformal risk control
$\hat{\lambda}$: $(n/(n+1))\hat{R} + 1/(n+1) \leq \alpha_{user}$

DENY
violation

approved

Tool call
AgentDojo environment

per-attack G, C, F

per-attack G, C, F

per-attack G, C, F

Output guard
canary token check + Qwen3Guard

Horizon risk + Sion game
loss $\ell = G + a \cdot C + h \cdot F$ · Sion LP → π^* (Horizon), w*
$V\ (\text{Horizon}) = \min\{1, \Sigma\pi\ q/|K| + \max(E[L] + \sqrt{(2\rho\ \text{Var}\ L)}, \Sigma\pi\ \ell\ adp)\}$

Answer

Legend

LLM / spotlighting

Math tool

Code / data

Direct / jailbreak layer
not in the Horizon math